

# DRG Virome Pipeline Update

Michael Wilde · CAPS · April 2026

**Background**

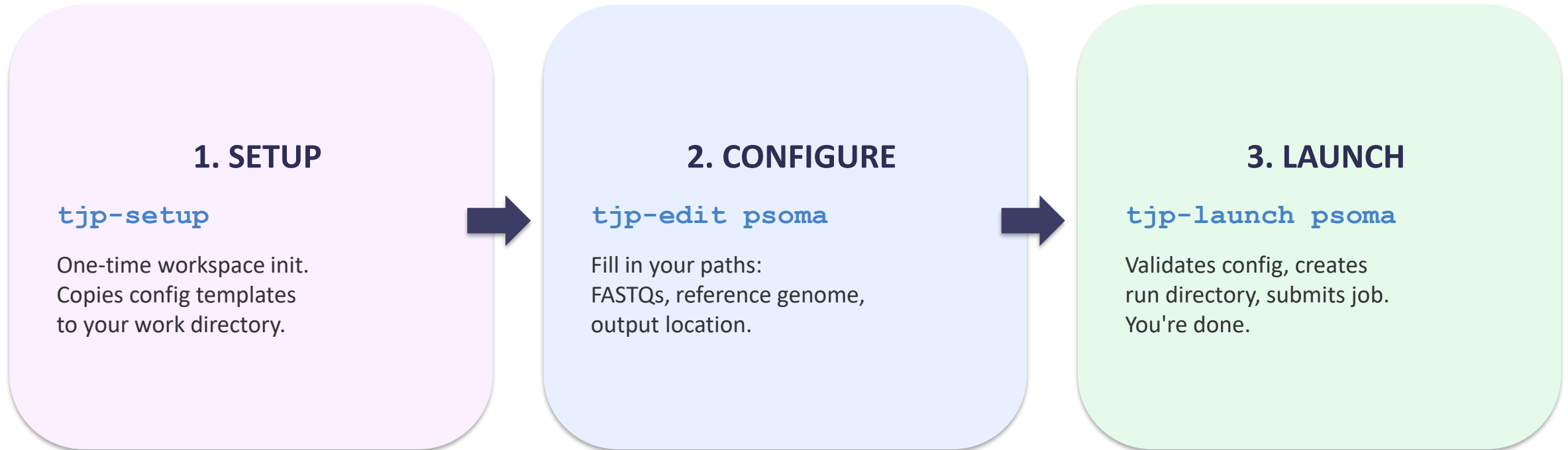
**Pipeline**

**PD cohort results**

**Next steps**

# HPC Pipeline Framework

Three commands. Full reproducibility. Zero SLURM knowledge required.



Every run is automatically tracked: config snapshot, reproducibility manifest, archived inputs & outputs.

**USER WORKFLOW:**      tjp-setup --> tjp-test --> tjp-test-validate --> tjp-launch --> tjp-validate  
                          (aliases: hyperion-\*, biocruiser-\*)

**templates/      CONFIG LAYER**

addone/config.yaml

bulktnaseq/config.yaml

cellranger/config.yaml

psoma/config.yaml

spaceranger/config.yaml

xeniumranger/config.yaml

Placeholders:    USER      SCRATCH      WORK      |      User configs: /work/\$USER/pipelines/<name>/config.yaml

**bin/      CLI LAYER**

tjp-setup: init workspace

tjp-test: run smoke test

tjp-test-validate: check test

tjp-launch: submit pipeline

tjp-validate: check out

lib/      common.sh (registry, paths)

validate.sh (per-pipeline)

manifest.sh (manifest.json)

branding.sh (themed output)

↓ sbatch

**slurm\_templates/      SLURM LAYER**

addone\_slurm\_template.sh

bulktnaseq\_slurm\_template.sh

psoma\_slurm\_template.sh

cellranger\_slurm\_template.sh

spaceranger\_slurm\_template.sh

xeniumranger\_slurm\_template.sh

**EXECUTION LAYER**

**CONTAINER PIPELINES (Apptainer)**

AddOne (.sif)  
inline

BulkRNASeq (.sif)  
submodule + ext NF

Psoma (.sif)  
submodule (combined)

Flags: --cleanenv    --env PYTHONNOUSERSITE=1  
Binds: PROJECT\_ROOT, SCRATCH\_ROOT, WORK\_ROOT

**NATIVE PIPELINES (no container)**

Cell Ranger 10.0.0  
/groups/tprice/opt/

Space Ranger 4.0.1  
/groups/tprice/opt/

Xenium Ranger 4.0  
/groups/tprice/opt/

Wrappers: containers/10x/ submodule (v1.1.0)  
Flags: --exclusive, --localcores, --localmem  
Config: tool\_path override supported

**HPC PATH LAYOUT**

/groups/tprice/pipelines/    shared repo

/groups/tprice/opt/          native tools

/work/\$USER/pipelines/      user configs & runs

/scratch/juno/\$USER/        execution & outputs

**RUN DIRECTORY (per launch)**

/work/\$USER/pipelines/<pipeline>/runs/<timestamp>/

config.yaml | manifest.json | slurm logs

inputs/ (archived FASTQs) | outputs/ (results)

# HPC Pipeline Framework

Three commands. Full reproducibility. Zero SLURM knowledge required.

## 1. SETUP

`tjp-setup`

One-time workspace init.  
Copies config templates  
to your work directory.



## 2. CONFIGURE

`tjp-edit psoma`

Fill in your paths:  
FASTQs, reference genome,  
output location.



## 3. LAUNCH

`tjp-launch psoma`

Validates config, creates  
run directory, submits job.  
You're done.

Every run is automatically tracked: config snapshot, reproducibility manifest, archived inputs & outputs.

# Viral Expression in DRG

## The question

- ▶ Neurotropic viruses (HSV-1, VZV, CMV) establish latency in DRG
- ▶ Viral reactivation implicated in neuropathic pain, neurodegeneration (ALS, MS)
- ▶ HSV-1 linked to Parkinson's in epidemiological studies
- ▶ No systematic virome profiling of human DRG has been reported

## The approach

- ▶ Use existing bulk RNA-seq host-deplete with STAR, classify with Kraken2
- ▶ Dual-database classification: viral-only vs. PlusPF (full taxonomy)
- ▶ Only taxa detected in BOTH databases = Tier 1 (confirmed)
- ▶ Baseline cohort (16 non-PD donors): Tier 1 = 0 (null validation)

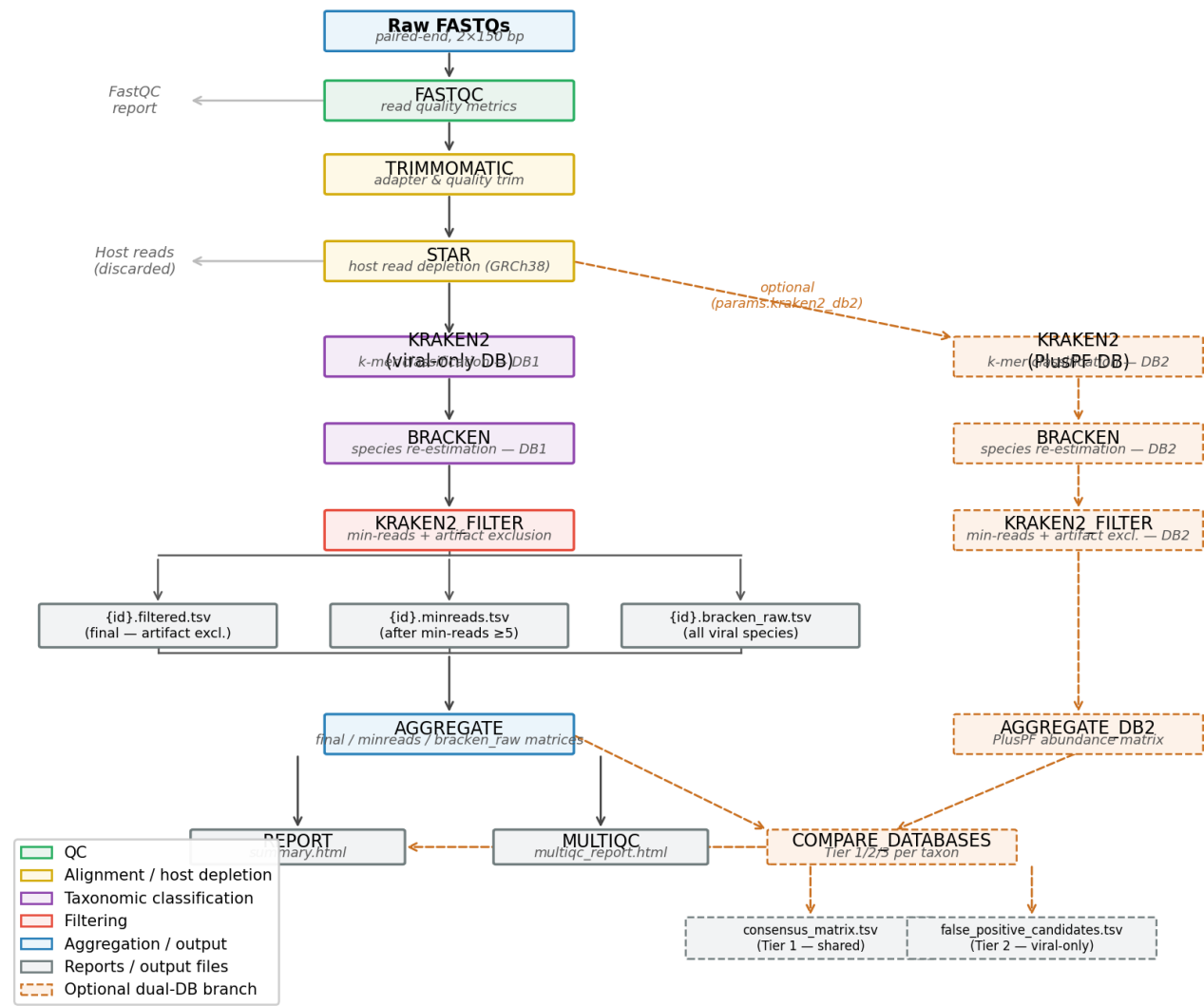
## What's new: Parkinson's DRG cohort

- ▶ 20 new DRG samples: 14 PD patients + 6 unclassified controls
- ▶ Same pipeline, same dual-database classification — first application to a disease cohort
- ▶ Result: the first Tier 1 detection across all 36 donors profiled to date

# Virus Detection Pipeline architecture — v1.5.0

7-step Nextflow DSL2 workflow + dual-database branch + BLAST verification offshoot

Figure 1 — virome-pipeline Data Flow (v1.3.0)



All steps containerized (Apptainer) | SLURM-ready | Each process in its own .sif container | stageInMode = 'copy' for Apptainer symlink compatibility

# The core innovation: dual-database competitive classification

*Run every sample through two Kraken2 databases simultaneously — only signals surviving both are real*

## Database 1: Viral-only

- ▶ ~10,000 viral genomes
- ▶ No non-viral sequences
- ▶ Closed-world: every read forced into a viral bin
- ▶ Maximizes sensitivity
- ▶ But: 100% FP rate in baseline cohort

## Three-tier confidence scoring

### Tier 1 — Shared

HSV-1 in PD19

Detected in BOTH databases  
High-confidence viral candidate  
Use for biology

### Tier 2 — Viral-only

HERV-K, CMV proxy, MCV

Viral-only DB exclusive  
False positive candidates  
Investigate before interpreting

### Tier 3 — PlusPF only

10–826 taxa / sample

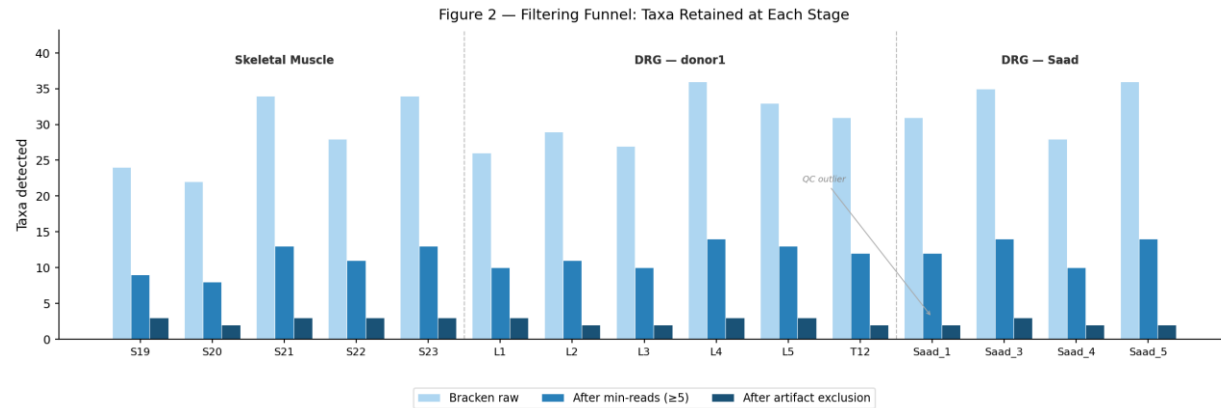
PlusPF-only  
Non-viral contaminants  
Quality Control

## Database 2: PlusPF (standard)

- ▶ Full bacterial, archaeal, viral, fungal + human
- ▶ ~100 GB, human genome included
- ▶ Open-world: reads compete against host
- ▶ HERV-K → reclassified to Homo sapiens
- ▶ CMV proxy → looking like an artifact

# Multi-stage filtering + curated artifact exclusion

*Bracken raw* → *min-reads threshold (≥5)* → *24-entry artifact exclusion list*



## Three output matrices per cohort

### **bracken\_raw**

All viral species from Bracken  
No threshold as baseline for QC

### **minreads**

≥5 reads per taxon per sample

### **final (filtered)**

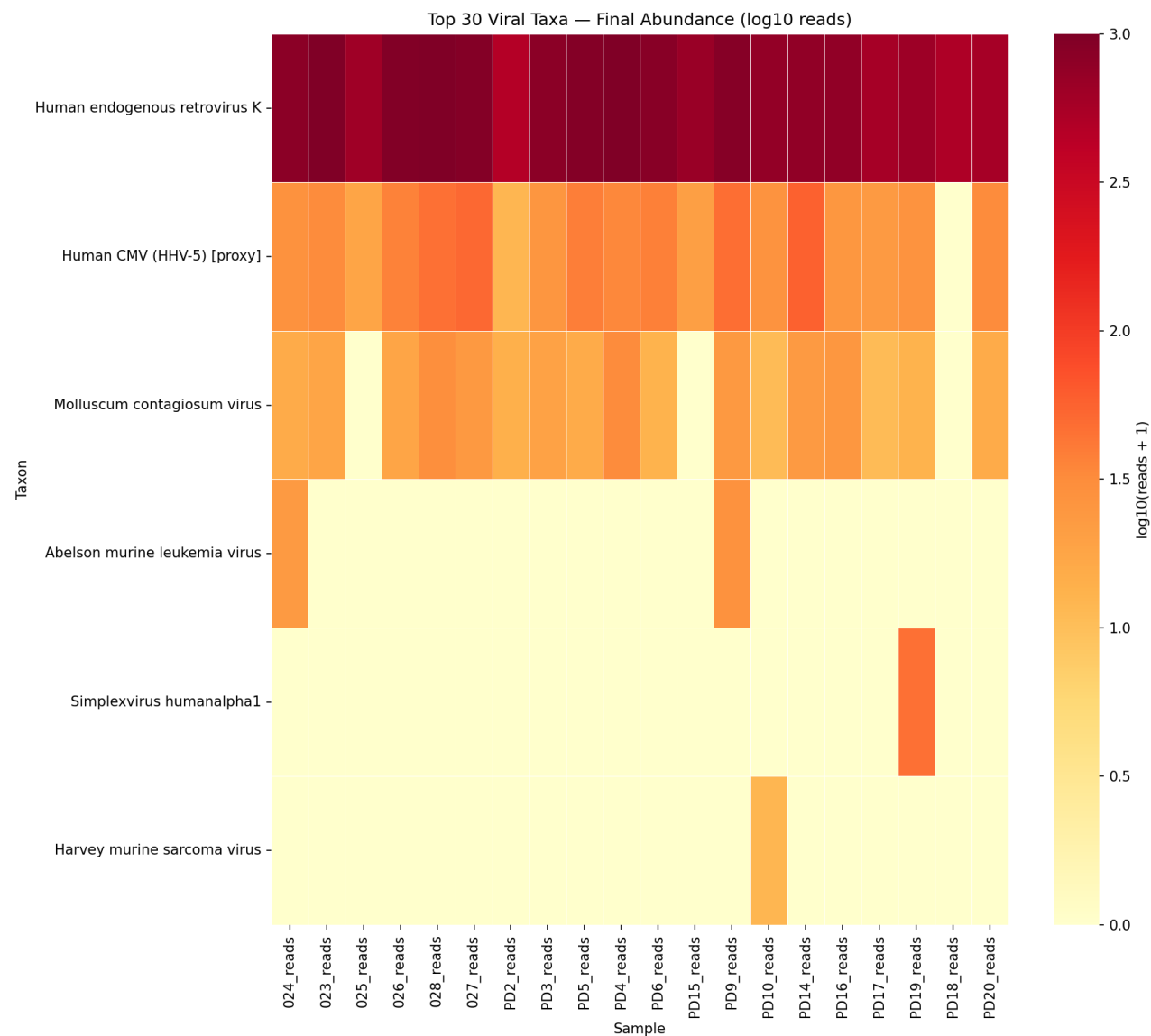
Curated artifact exclusion applied  
24 taxa: phages, reagent contaminants, env. metagenome, cross-mapping

## Artifact categories

Reagent contaminants · ruminant orthobunyaviruses · insect baculoviruses · phages (ICTV-reclassified) · DRG-specific env. metagenome cross-mapping · avian herpesviruses · giant amoeba viruses · hantaviruses (Oxbow)



# Viral abundance heatmap — PD cohort (n=20)

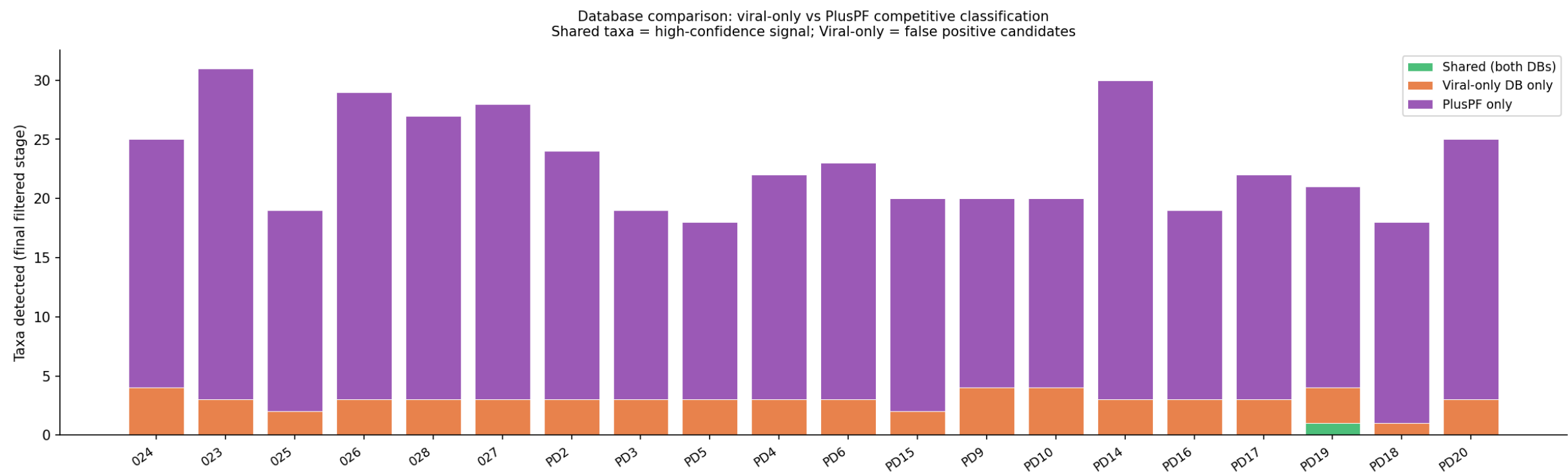


## Key observations

- ▶ HERV-K dominant in all samples (endogenous, not infection)
- ▶ CMV proxy present but Tier 2 (k-mer artifact)
- ▶ MCV sporadic (index hopping)
- ▶ Simplexvirus humanalpha1 = HSV-1
- ▶ HSV-1 detected ONLY in PD19
- ▶ 46 reads, 1.89 RPM
- ▶ Absent from all other 19 samples

HSV-1 in PD19 is the only Tier 1 detection across all 36 donors

# Dual-database tier classification — PD cohort



**Shared (Tier 1)**

Detected in BOTH DBs = confirmed viral signal

1 taxon in 1 sample (HSV-1, PD19)



**Viral-only (Tier 2)**

Viral-only DB exclusive = false positive candidates

3 taxa per sample (HERV-K, CMV proxy, MCV)



**PlusPF only (Tier 3)**

Non-viral background contaminants

15–30 taxa per sample (QC utility)

PD19 is the only sample with a green (Shared) segment | Baseline cohort (n=16 non-PD donors) had zero Tier 1 detections

# PD19 HSV-1

46

reads (Kraken2)

1.89

RPM

1/36

donors with HSV-1

0/22

non-PD with HSV-1

1/14

PD with HSV-1

## Why this is plausible

- ▶ HSV-1 naturally establishes latency in sensory ganglia
- ▶ Antiviral therapy linked to reduced PD risk in registries (UK BioBank)
- ▶ Proposed mechanisms: retrograde transport,  $\alpha$ -synuclein interaction, neuroinflammation
- ▶ 46 reads well above ~10 read noise floor
- ▶ Survived competitive classification against full human genome

## What's pending

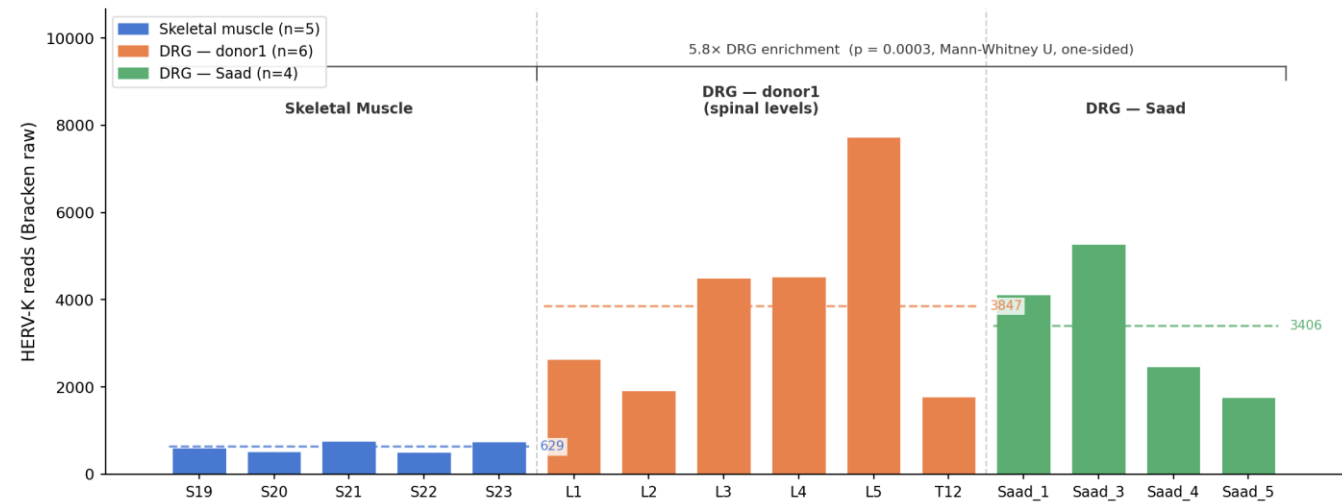
- ▶ BLAST verification (blast\_verify.nf ready to run)
- ▶ Life cycle phase: latency (LAT) or reactivation (IE/E/L)?
- ▶ Genome coverage profile against NC\_001806.2
- ▶ Not statistically significant yet (Fisher's  $p \approx 0.35$ ,  $n=1$ )

**This is a hypothesis, not a conclusion. BLAST validation and a larger cohort are required before any biological claim.**

# HERV-K — a real biological signal worth tracking

5.8× DRG enrichment over muscle · reproduced across cohorts · first quantification of HERV-K in human DRG

Figure 4 — HERV-K Endogenous Retroviral Signal by Tissue Type and Spinal Level



## Cross-cohort HERV-K comparison

|                   |      |          |
|-------------------|------|----------|
| Muscle (non-PD)   | n=5  | 11.3 RPM |
| DRG — donor1      | n=6  | 45.8 RPM |
| DRG — Saad        | n=4  | 72.7 RPM |
| DRG — PD patients | n=14 | 31.8 RPM |
| DRG — PD controls | n=6  | 34.5 RPM |

## Implications

- ▶ HERV-K is NOT a virus — it's endogenous retroviral chromosomal transcription
- ▶ DRG enrichment likely reflects neural-lineage chromatin accessibility
- ▶ ALS literature: HERV-K Env protein in motor neurons (Li et al. 2015)
- ▶ PD de-repression hypothesis NOT supported (PD RPM ≤ non-PD RPM)
- ▶ Could serve as internal control — HERV-K RPM as library quality metric

# Next steps

## This month

- ▶ BLAST validation of PD19 HSV-1
- ▶ Determine latency vs. reactivation
- ▶ Begin compiling and cataloguing bulk seq

## Soon

- ▶ Begin lab-wide bulk RNA-seq inventory
- ▶ Compile samplesheets for existing data
- ▶ Run across conditions (MS, COVID, etc)

## Longer term

- ▶ 100+ donor DRG virome atlas
- ▶ conditions comparison paper
- ▶ Expand to snRNA-seq / spatial (pseudobulk)

## The ask

- ▶ I'll be inventorying existing bulk RNA-seq (and all other sequencing data): which donors, tissues, and FASTQs are available?

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Questions?

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